CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Total number of sigma (σ) _____ and pi(π) _____ bonds respectively present in hex-1-en-4-yne are :

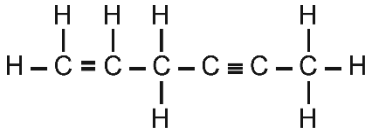
(1) 3 and 13

(2) 11 and 3

(3) 14 and 3

(4) 13 and 3

Answer (4)

Sol. 

σ bond = 13, π bond = 3

52. Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16 mg/mL, which becomes 4 mg/mL, in 12 months. The expiry time of the drug in months is _____

Assume that decomposition of drug follows first order kinetics

(1) 6

(2) 12

(3) 3

(4) 2

Answer (1)

Sol. Drug X $\xrightarrow{\text{1st order}}$ Product

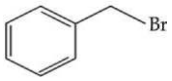
Initial concentration of drug = 16 mg/mL

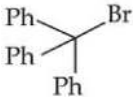
Concentration of drug after 12 months = 4 mg/mL

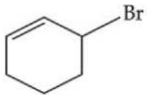
Half life of drug = 6 months

Drug become ineffective after 50% decomposition. The expiry time of drug = 6 months

53. Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?

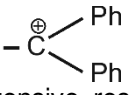
(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol.  is most stable carbocation due to extensive resonance stabilization from the three phenyl groups.

54. 0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation

$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the **correct** statements :

(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3

(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4

(C) 0.5 L of KI solution produced 0.005 mol of I_2

(D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$

Choose the **correct** answer from the options given below :

(1) (A) and (D) only (2) (C) and (D) only

(3) (A) and (B) only (4) (B) and (C) only

Answer (1)

Sol. $E_{KIO_3} = \frac{\text{Molecular weight}}{n_f}$

$n_f = 5$

$E_{KIO_3} = \frac{\text{Molecular weight}}{5}$

(D) is correct

meq of KI = $0.1 \times 200 = 20$

meq of $KIO_3 = 4 \times 5 = 20$

(A) is correct



55. Identify the homoleptic complexes with odd number of d electrons in the central metal :

(A) $[FeO_4]^{2-}$

(B) $[Fe(CN)_6]^{3-}$

(C) $[Fe(CN)_5NO]^{2-}$

(D) $[CoCl_4]^{2-}$

(E) $[Co(H_2O)_3F_3]$

Choose the **correct** answer from the options given below :

(1) (A), (B) and (D) only

(2) (A), (C) and (E) only

(3) (C) and (E) only

(4) (B) and (D) only

Answer (4)

Sol. (A) $[FeO_4]^{2-} \Rightarrow Fe^{6+} = 3d^2$

(B) $[Fe(CN)_6]^{3-} \Rightarrow Fe^{3+} = 3d^5$

(C) $[Fe(CN)_5NO]^{2-} \Rightarrow Fe^{2+} = 3d^6$

(D) $[CoCl_4]^{2-} \Rightarrow Co^{2+} = 3d^7$

(E) $[Co(H_2O)_3F_3] \Rightarrow Co^{3+} = 3d^6$

(B) and (D) are homoleptic complex having odd no. of d electrons.

56. First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

(1) 1012 kJ mol^{-1}

(2) 834 kJ mol^{-1}

(3) 1402 kJ mol^{-1}

(4) 947 kJ mol^{-1}

Answer (1)

Sol. The main component of apatite family is phosphorus.

Order of IE_1 of group 15 elements N > P > As > Sb

IE of phosphorus = 1012 kJ mol^{-1}.

57. If C(diamond) $\rightarrow$ C(graphite) + X kJ mol^{-1}

C(diamond) + $O_2(g) \rightarrow CO_2(g)$ + Y kJ mol^{-1}

C(graphite) + $O_2(g) \rightarrow CO_2(g)$ + Z kJ mol^{-1}

at constant temperature. Then

(1) X = –Y + Z (2) X = Y – Z

(3) X = Y + Z (4) –X = Y + Z

Answer (2)

Sol. C(diamond) + $O_2(g) \rightarrow CO_2(g)$; $\Delta H_1 = -Y$ kJ mol^{-1}

$CO_2(g) \rightarrow$ C(graphite) + $O_2(g)$; $\Delta H_2 = Z$ kJ mol^{-1}

C(diamond) $\rightarrow$ C(graphite) ; $\Delta H_3 = -Y + Z$

– X = – Y + Z

X = Y – Z

58. Given below are two statements :

Statement-I : On nitration of m-xylene with HNO_3, H_2SO_4 followed by oxidation, 4-nitrobenzene-1,3-dicarboxylic acid is obtained as the major product.

Statement-II : $-CH_3$ group is o/p-directing while $-NO_2$ group is m-directing group.

In the light of the above statement, choose the correct answer from the options given below:

(1) Both **Statement-I** and **Statement-II** are false

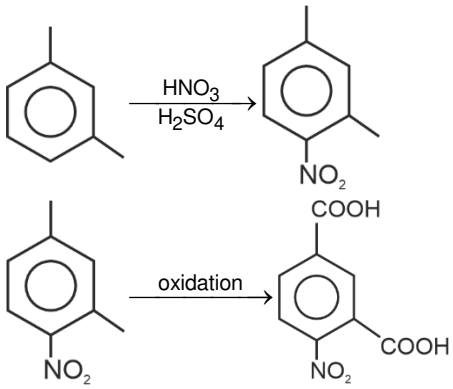
(2) Both **Statement-I** and **Statement-II** are true

(3) **Statement-I** is false but **Statement-II** is true

(4) **Statement-I** is true but **Statement-II** is false

Answer (2)

Sol.



$-CH_3$ group is o/p directing group and

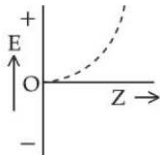
$-NO_2$ is m-directing group.

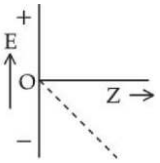


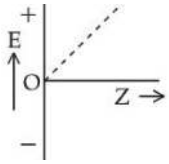
59. For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n ?

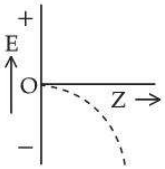
[E : Energy of the stationary state,

Z : atomic number, n = principal quantum number]

(1) 

(2) 

(3) 

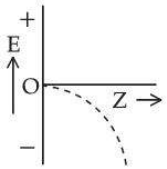
(4) 

Answer (4)

Sol. $E_n = -13.6\frac{z^2}{n^2}$

$E_n \propto -z^2$

$y = kx^2$



60. Match **List-I** with **List-II :**

	List-I Applications		**List-II Batteries/Cell**
(A)	Transistors	(I)	Anode – Zn/Hg; Cathode – HgO + C
(B)	Hearing aids	(II)	Hydrogen fuel cell
(C)	Invertors	(III)	Anode – Zn; Cathode - Carbon
(D)	Apollo space ship	(IV)	Anode – Pb; Cathode – Pb | PbO_2

Choose the **correct** answer from the options given below :

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Answer (4)

Sol. In transistor, anode is Zn and cathode is carbon.

In hearing, aids mercury battery are used.

In invertors, lead storage battery is used.

In apollo space ship, hydrogen fuel cell was used.

61. Given below are two statements :

Statement (I) : It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II) : If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is false but **Statement-II** is true

(2) Both **Statement-I** and **Statement-II** are true

(3) Both **Statement-I** and **Statement-II** are false

(4) **Statement-I** is true but **Statement-II** is false

Answer (2)

Sol. According to Heisenberg's uncertainty principle, it is impossible to determine simultaneously the exact position and momentum of particle like electron

If $\Delta p = \Delta x$

$$\Delta p.\Delta x \geq \frac{h}{4\pi}$$

$$(\Delta p)^2 \geq \frac{h}{4\pi}$$

$$\Delta p \geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2}$$

$$m\Delta v \geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2}$$

$$\Delta v \geq \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$$



62. Given below are two statements :

Statement (I) : In partition chromatography, stationary phase is thin film of liquid present in the inert support.

Statement (II) : In paper chromatography, the material of paper acts as a stationary phase.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is true but **Statement-II** is false

(2) Both **Statement-I** and **Statement-II** are false

(3) **Statement-I** is false but **Statement-II** is true

(4) Both **Statement-I** and **Statement-II** are true

Answer (1)

Sol. Paper chromatography is a type of position chromatography in which a special quality paper is used known as chromatography paper, chromatography paper contains water trapped in it, which acts as stationary phase.

63. Given below are two statements :

Statement (I) : NaCl is added to the ice at 0°C, present in the ice cream box to prevent the melting of ice cream.

Statement (II) : On addition of NaCl to ice at 0°C, there is a depression in freezing point.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement-I** is true but **Statement-II** is false

(2) Both **Statement-I** and **Statement-II** are true

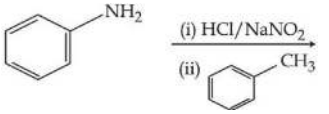
(3) **Statement-I** is false but **Statement-II** is true

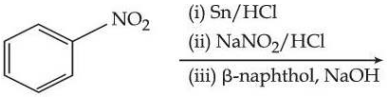
(4) Both **Statement-I** and **Statement-II** are false

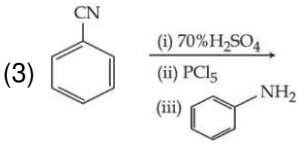
Answer (2)

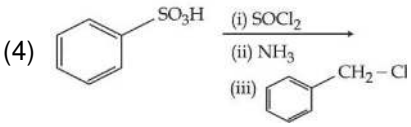
Sol. A mixture of salt and ice is known as freezing mixture. Freezing mixture decreases freezing point of ice. Both statements are true.

64. Which one of the following reaction sequences will give an azo dye?

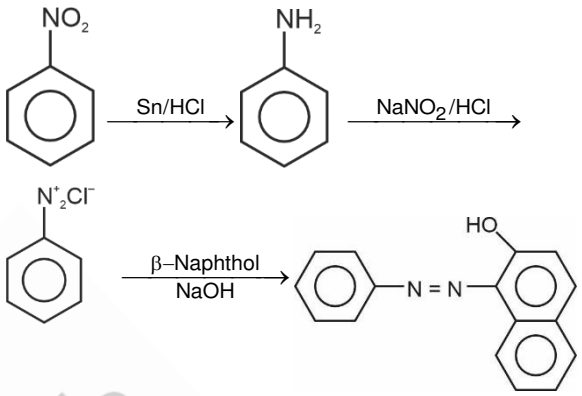
(1) 

(2) 

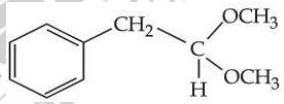
(3) 

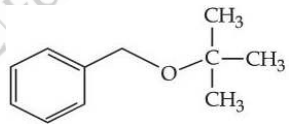
(4) 

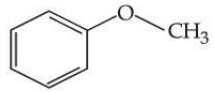
Answer (2)

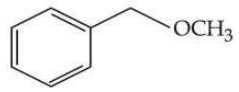
Sol. 

65. Which one of the following, with HBr will give a phenol?

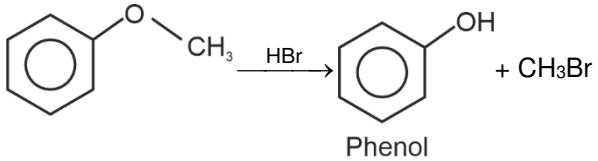
(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. 



66. The calculated spin-only magnetic moments of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are :

(1) 3.87 and 4.90 B.M.

(2) 5.92 and 4.90 B.M.

(3) 4.90 and 4.90 B.M.

(4) 4.90 and 5.92 B.M.

Answer (2)

Sol.

$K_3\left[Fe(OH)_6\right]$

$Fe^{3+} \Rightarrow 3d^5$

Fe^{3+} with OH^- (WFL)

$= t_{2g}^3 e_g^2$

Number of unpaired electron (n) = 5

$\mu_{spin\ only}$ = 5.92 BM

$K_4\left[Fe(OH)_6\right]$

$Fe^{2+} \Rightarrow OH^-$ WFL

$Fe^{2+} \Rightarrow 3d^6 = t_{2g}^4\ e_g^2$

n = 4

$\mu_{spin\ only}$ = 4.90 BM

67. Identify the essential amino acids from below :

(A) Valine

(B) Proline

(C) Lysine

(D) Threonine

(E) Tyrosine

Choose the **correct** answer from the options given below :

(1) (B), (C) and (E) only

(2) (C), (D) and (E) only

(3) (A), (C) and (E) only

(4) (A), (C) and (D) only

Answer (4)

Sol. Valine, Lysine and Threonine are example of essential amino acid.

68. Consider the equilibrium

$$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$$

If the pressure applied over the system increases by two fold at constant temperature then

(A) Concentration of reactants and products increases.

(B) Equilibrium will shift in forward direction.

(C) Equilibrium constant increases since concentration of products increases.

(D) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the **correct** answer from the options given below :

(1) (B) and (C) only

(2) (A), (B) and (C) only

(3) (A) and (B) only

(4) (A), (B) and (D) only

Answer (3)

Sol. $CO(g) + 3H_2 \rightleftharpoons CH_4(g) + H_2O(g)$

$\Delta n_g = -2$

If pressure of system increases then according to Le-Chatelier's principle reaction will move in forward direction.

Concentration of reactant and products both increases but concentration of product increases more.

69. O_2 gas will be evolved as a product of electrolysis of :

(A) An aqueous solution of $AgNO_3$ using silver electrodes.

(B) An aqueous solution of $AgNO_3$ using platinum electrodes.

(C) A dilute solution of H_2SO_4 using platinum electrodes.

(D) A high concentration solution of H_2SO_4 using platinum electrodes.

Choose the **correct** answer from the options given below :

(1) (A) and (D) only (2) (B) and (D) only

(3) (B) and (C) only (4) (A) and (C) only

Answer (3)



Sol. When an aqueous solution of $AgNO_3$ is electrolysed using Pt electrodes

Cathode : $Ag^+(aq) + e^- \rightarrow Ag(s)$

Anode : $2H_2O(l) \rightleftharpoons 4H^+(aq) + O_2(g) + 4e^-$

$\Rightarrow$ When dilute H_2SO_4 is electrolysed using Pt electrodes

Anode : $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$

Cathode : $2H^+(aq) + 2e^- \rightarrow H_2(g)$

70. The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

(1) A_2O (2) AO

(3) A_2O_3 (4) AO_2

Answer (3)

Sol. Among given atoms, Boron has least atomic radius oxide of Boron = B_2O_3

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

71. Isomeric hydrocarbons $\rightarrow$ negative Baeyer's test

(Molecular formula C_9H_{12})

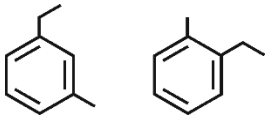
The total number of isomers from above with four different non-aliphatic substitution site is –

Answer (2)

Sol. Degree of unsaturation = $C + 1 - \frac{H}{2}$

$= 9 + 1 - 6 = 4$

Benzene shows negative Baeyer's test



Both compounds have four different non-aliphatic substitution sites.

72. Total number of non-bonded electrons present in NO_2^- ion based on Lewis theory is _____.

Answer (12)

Sol. $\ddot{O} = \ddot{N} - \ddot{O}:^-$

Number of non-bonding electrons

= 12

73. In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is _____ $\times 10^{-1}$%.

[Molar mass : O = 16, S = 32, Ba = 137 in g mol^{-1})

Answer (275)

Sol. Moles of $BaSO_4$ = Moles of S = $\frac{0.40}{233}$ mol

Mass of S = $\frac{0.40}{233} \times 32$ g

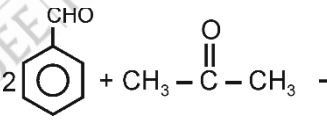
= 0.055 g

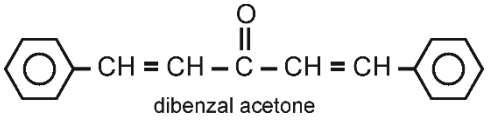
%S = $\frac{0.055}{0.20} \times 100 = 27.5\%$

= 275×10^{-1} %

74. In the Claisen-Schmidt reaction to prepare, dibenzalacetone from 5.3 g of benzaldehyde, a total of 3.51 g of product was obtained. The percentage yield in this reaction was _____ %.

Answer (60)

Sol. 



dibenzal acetone

Mole = $\frac{5.3}{106} = 0.05$ mol

2 mol benzaldehyde produces 1 mol dibenzalacetone

moles of dibenzalacetone = 0.025 mol

Mass = 0.025×234 g = 5.85 g

% yield = $\frac{3.51}{5.85} \times 100$

= 60 %



75. Consider the following low-spin complexes

$K_3[Co(NO_2)_6]$, $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $Cu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$.

The sum of the spin-only magnetic moment values of complexes having yellow colour is _____ B.M. (answer in nearest integer)

Answer (0)

Sol. $K_3[Co(NO_2)_6]$ = Yellow $\Rightarrow$ $Co^{3+} = 3d^6 \Rightarrow t_{2g}^6 e_g^0$

$K_4[Fe(CN)_6]$ = Yellow $\Rightarrow$ $Fe^{2+} = 3d^6 \Rightarrow t_{2g}^6 e_g^0$

$K_3[Fe(CN)_6]$ = Bright Red $\Rightarrow$ $Fe^{3+} = 3d^5 \Rightarrow t_{2g}^5 e_g^0$

$Cu_2[Fe(CN)_6]$ = Chocolate brown

$Zn_2[Fe(CN)_6]$ = White

Spin only magnetic moment of complex having Yellow colour is zero.

❑ ❑ ❑

